# Akoizo Original Practice - Form 4 - Answer key

### Division C | Wright Stuff | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

### ANSWER KEY - Review only after submitting.

#### 1. (2 points) What is propeller pitch?

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

A. The blade thickness

B. The number of blades

C. The ideal distance advanced in one revolution

D. The tip-to-tip span

**Correct: C**

#### 2. (2 points) Which change lowers the flying speed of a plane of fixed weight?

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

A. Adding weight

B. Reducing the lift coefficient

C. Increasing wing area

D. Decreasing wing area

**Correct: C**

**3. (2 points) What does high remaining torque at landing suggest?**

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

A. The plane descended before the motor ran down; trim or propeller may need changing

B. The motor was too weak

C. The flight was perfect

D. The ceiling was too high

**Correct: A**

**4. (2 points) Why do wing dihedral and a fin matter in a circling plane?**

Unit: Stability, trim, and circling flight.
Worked-case data: A plane stalls on every glide test. List adjustments in order to try.
Separate investigation observations:
Glide: Smooth and straight.
Powered launch: Plane rolls left and spirals down in the first 5 s.
After 5 s: Torque drops; plane recovers if still airborne.

A. They prevent the turn from tightening into a spiral dive

B. They increase thrust

C. They reduce mass

D. They add lift only

**Correct: A**

**5. (4 points) Describe the three parts of a rubber motor’s torque curve.**

Unit: Rubber motors: storing and releasing energy.
Worked-case data: A team uses a 15:1 winder and tests show a motor breaks near 1,400 turns. How many cranks should they wind for a safe 8 percent back-off?
Separate investigation observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

answer: A high initial burst, a long, fairly flat cruise portion, and a tail-off as the last turns unwind. (4 points).

**6. (4 points) List the three required flight log parameters.**

Unit: Rules, scoring, and the flight log bonus.
Worked-case data: A team with a complete flight log flies official flights of 96.4 s and 112.0 s. What is its score and tiebreaker?
Separate investigation observations:
Team A: Best flight 120 s, no flight log.
Team B: Best flight 105 s, complete flight log.
Team C: Best flight 112 s, partial flight log.

answer: Motor size before wind-up; turns or torque at launch; flight time. (4 points).

**7. (4 points) What can warp a plane between practice and competition?**

Unit: Building light and strong.
Worked-case data: A built airframe weighs 7.4 g. The CG is slightly behind the target. How should the team meet the 8.0 g minimum?
Separate investigation observations:
Flight 1: Fresh build, 102 s.
Flight 6: After a week in a hot car, 61 s; plane turned sharply right.
Inspection: Right wing twisted 2 degrees.

answer: Heat, humidity, and rough handling can twist balsa and film. (4 points).

**8. (4 points) What is slip, and why does it exist?**

Unit: Propellers: turning torque into thrust.
Worked-case data: A plane cruises at 1.2 m/s and its propeller effectively advances 0.24 m per turn. A motor has 1,200 usable turns. Estimate motor run time.
Separate investigation observations:
Prop 1: Diameter 22 cm; plane cruised level; motor ran 140 s.
Prop 2: Diameter 26 cm, same pitch; plane sank after the burst; 90 s.
Prop 3: Diameter 18 cm; plane climbed fast; motor ran 95 s.

answer: The difference between pitch and actual advance; the blades must meet the air at an angle to make thrust. (4 points).

**9. (4 points) Why should drafts be noted in the flight log?**

Unit: Wings and slow flight.
Worked-case data: A 9.5 g plane (with motor) has 0.040 m^2 of wing and C_L = 0.8. Find its level-flight speed (ρ = 1.2 kg/m^3).
Separate investigation observations:
Plane 1: Wing area 0.030 m^2; cruise 2.6 m/s; flight 85 s.
Plane 2: Wing area 0.042 m^2; cruise 2.1 m/s; flight 118 s.
Same: Mass, motor, and turns.

answer: Drafts push slow planes off course and change flight times; noting them explains outliers and guides circle placement. (4 points).

**10. (4 points) What is a hang-up and how can teams avoid it?**

Unit: Testing, the flight log, and competition day.
Worked-case data: Log data at a 9 m ceiling: torque 0.10 -> 95 s, 0.13 -> 118 s, 0.16 -> 104 s (ceiling bumps). Plan the first official flight.
Separate investigation observations:
Room A, 7 m ceiling: Best time at torque 0.10.
Room B, 11 m ceiling: Best time at torque 0.15.
Competition: Ceiling announced as 9 m.

answer: The plane catches on a beam or light and stops being supported by its wings; avoid by circling clear of fixtures and controlling the climb. (4 points).

**11. (6 points) Solve this case and explain your reasoning: A plane stalls on every glide test. List adjustments in order to try.**

reasoning: Check the CG; move it slightly forward with a tiny ballast if behind plan.
Reduce decalage by lowering the stabilizer's leading edge slightly.
Retest the glide and record the change and result. (3 points).

conclusion: Make one small change at a time until the glide is smooth. (3 points).

**12. (2 points) Why did Motor B fly longer?**

Supplied synthetic observations:
Motor A: 1.4 g, 30 cm long; 1,100 turns; climbed to ceiling, 78 s.
Motor B: 1.4 g, 38 cm long; 1,350 turns; stayed below ceiling, 104 s.
Ceiling: 8 m in both tests.

A. It was wound with fewer turns

B. Its lower torque and more turns gave a gentle climb and longer cruise

C. It had more mass

**Correct: B**

**13. (2 points) Define "Measurement box" in the context of Wright Stuff.**

answer: 32.0 x 24.0 x 47.0 cm; the plane must fit in flight configuration. (2 points).

**14. (2 points) Define "Spar" in the context of Wright Stuff.**

answer: Main lengthwise member of a wing. (2 points).

**15. (2 points) Define "Diameter" in the context of Wright Stuff.**

answer: Tip-to-tip span of the propeller. (2 points).

**16. (2 points) Define "Wing loading" in the context of Wright Stuff.**

answer: Weight divided by wing area. (2 points).

**17. (2 points) Define “Torque at landing” in the context of Wright Stuff.**

answer: Remaining motor torque when the plane lands. (2 points).

**18. (2 points) Define “Dihedral” in the context of Wright Stuff.**

answer: Upward wing angle that adds roll stability. (2 points).